Graphs of logarithmic functions

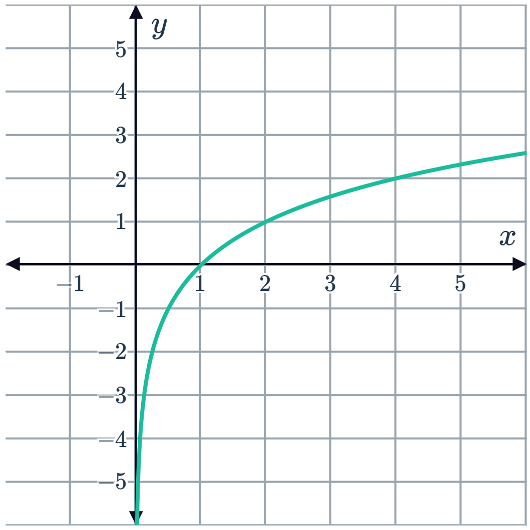
1 A

2

a

x	$\frac{1}{2}$	1	2	4	16
y	-1	0	1	2	4

b



c $x = 0$

3

a

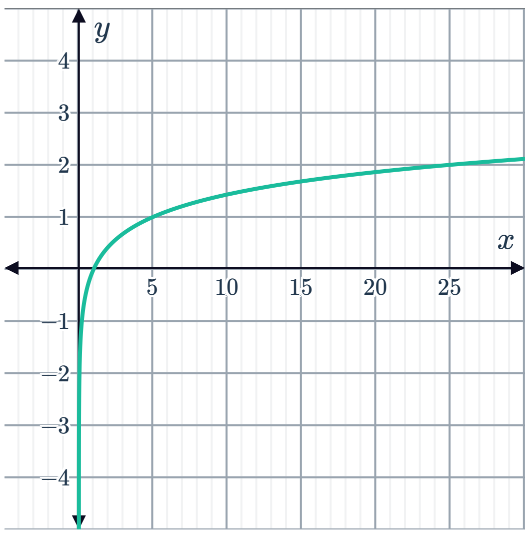
x	$\frac{1}{16}$	$\frac{1}{4}$	4	16	256
y	-2	-1	1	2	4

b $(1, 0)$

c 0

d $x = 4$

4



5

a

x	$\frac{1}{1024}$	$\frac{1}{4}$	1	4	16	256
y	-5	-1	0	1	2	4

b Increasing

c $\log_4 x$ approaches $-\infty$ as x approaches zero.

d y is undefined

6

a **B**

b **A**

c There is no value that $\log_a x$ will always be greater than.

d There is no value that $\log_a x$ will always be less than.

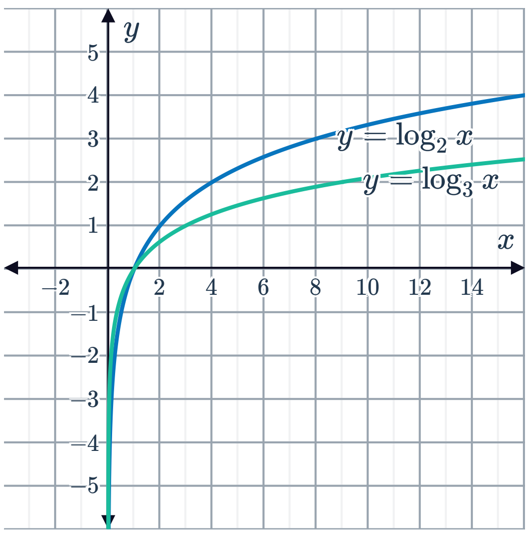
7

a $\log_a x$ is decreasing

b $\frac{2}{3}$

8

a



b The larger the base a of the function $y = \log_a x$, the less steep the graph.

9

a $k = 3$

b $f(x) = \log_3 x$

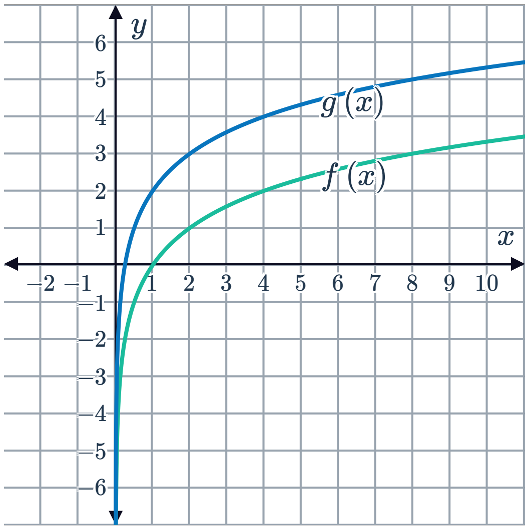
Transformations of logarithmic functions

10

a

x	$\frac{1}{2}$	1	2	4	8
$f(x) = \log_2 x$	-1	0	1	2	3
$g(x) = \log_2 x + 2$	1	2	3	4	5

b



c $g(x) = f(x) + 2$. A translation of 2 units upwards.

d

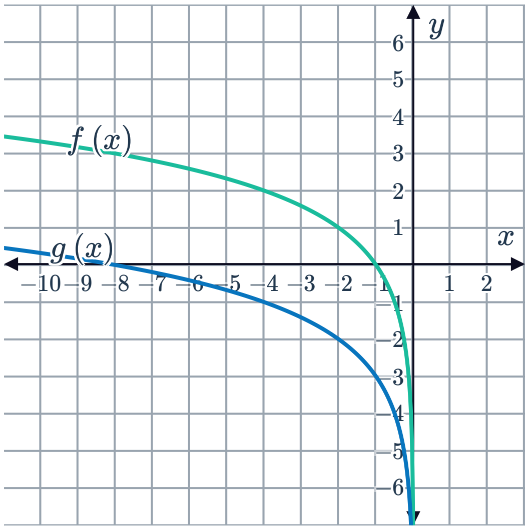
i Yes **ii** Yes **iii** No **iv** Yes

11

a

x	-8	-4	-2	-1	$-\frac{1}{2}$
$f(x) = \log_2(-x)$	3	2	1	0	-1
$g(x) = \log_2(-x) - 3$	0	-1	-2	-3	-4

b



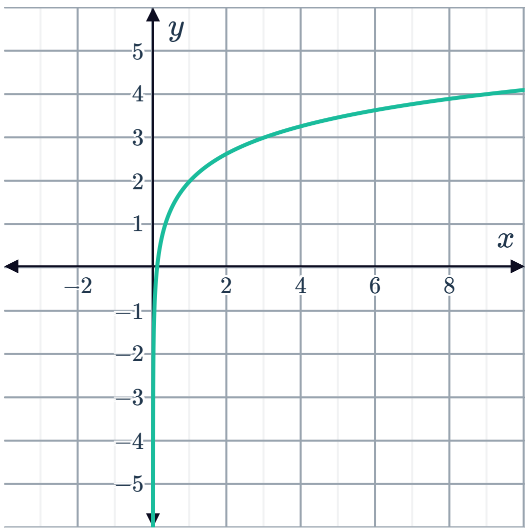
c $g(x) = f(x) - 3$. A translation of 3 units downwards.

d

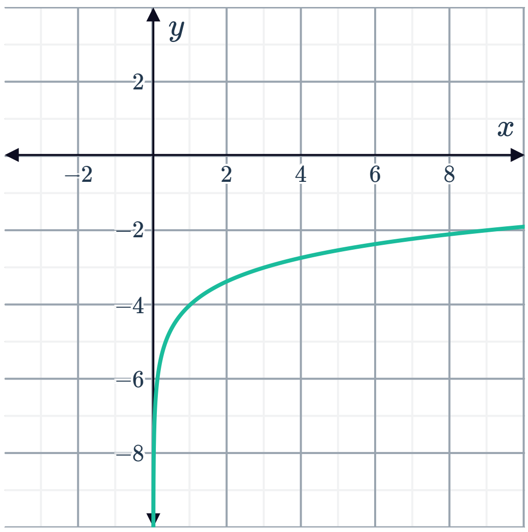
i Yes **ii** Yes **iii** No **iv** Yes

12

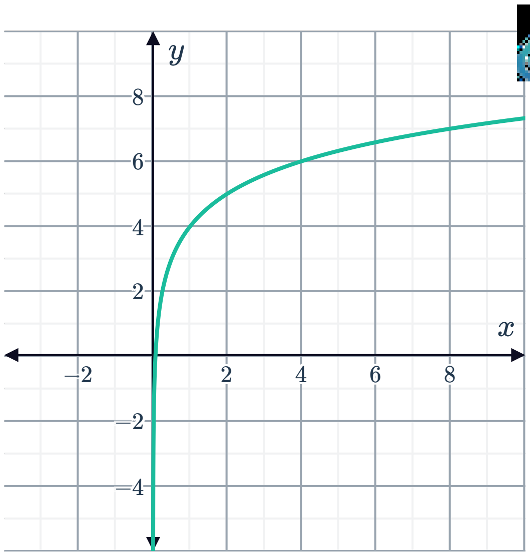
a



b



c



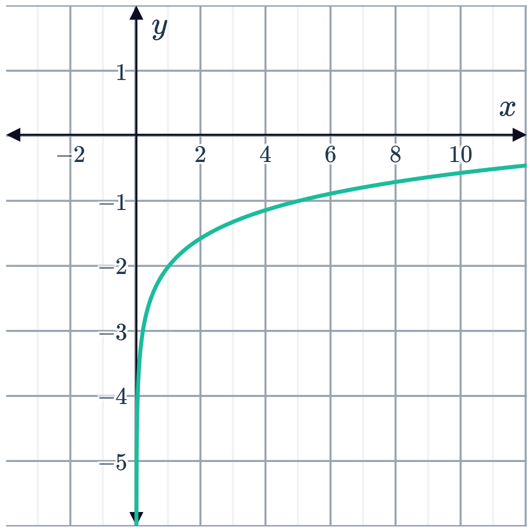
13 Translated upwards by 4 units

14

a

i $y = \log_5 x - 2$

ii



b

i $y = \log_3(-x) + 2$

ii